

General Features

Sensor	Internal 5Hz high sensitivity geophone Internal 10Hz high sensitivity geophone Optional KCK version for external string
Digitalization	32-bit Δ - Σ ADC with GNSS disciplined clock
Storage Capacity	16GB (32GB optional)
Node Autonomy	Internal 54/60WH lithium-ion battery pack supports 45/50 days continuous recording
Timing Accuracy	$\pm 10\mu s$ (GNSS disciplined) $\pm 1ms$ (within 1 hour without GNSS signal)
Automated Tests	Internal testing signal generator for function and performance self-inspection and diagnosis
Working Mode	Autonomy with wireless on-site QC inspection, Support UAV patrol
LED Status Indicator	Nodal health, GNSS synchronization status, Wireless data harvesting status
Data Harvest	Charging/Data download rack
Infield Management	Handheld android tablet
Weight	750g (excl. spike), 800g (incl. spike)
Size	9.8cm x 10.8cm x 11cm (excl. spike)
Operating Temperature	-40°C ~ +70°C

Acquisition Specifications

ADC Resolution	32-bit	
Sample Interval	0.5ms, 1ms, 2ms, 4ms	
Preamplifier Gain	x1, x2, x4, x8, x16, x32, x64 (0dB, 6dB, 12dB, 18dB, 24dB, 30dB, 36dB)	
Input Impedance	40K Ω 22nF (5Hz geophone) 20K Ω 22nF (10Hz geophone)	
Gain Accuracy	0.1%	
Full Scale Input Signal	$\pm 2.5Vp-p$ @ x1 gain $\pm 156mVp-p$ @ x16 gain	$\pm 625mVp-p$ @ x4 gain $\pm 39mVp-p$ @ x64 gain
Instantaneous DR @4ms (typical value)	125dB @ x1 gain 115dB @ x16 gain	122dB @ x4 gain 106dB @ x64 gain
Full Scale DR	142dB	
Equivalent Input Noise @4ms (typical value)	0.99 μV @ x1 gain 0.20 μV @ x16 gain	0.35 μV @ x4 gain 0.14 μV @ x64 gain
Harmonic Distortion	-120dB @ 31.5Hz	
Frequency Response @0.5ms	0Hz ~ 826Hz (-3dB)	
Anti-alias Filter	Linear or minimum phase filter, 82.6% of Nyquist -3dB bandwidth	
DC-blocking filter	0.1 ~ 10Hz	
Stopband Attenuation	>130dB @Nyquist	
Common Mode Rejection	>120dB	

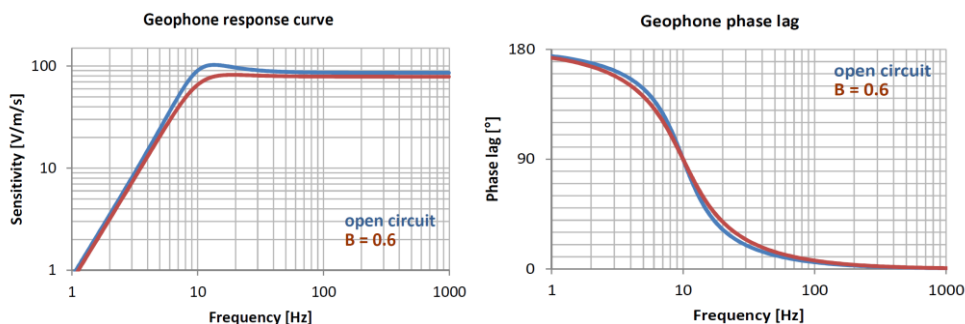
Product Highlights

- Built-in 5Hz (5Hz~150Hz) or 10Hz (10Hz~240Hz) high sensitivity geophone;
- Optional KCK version for external geophone string;
- 32-bit Δ - Σ ADC converter for high resolution, high dynamic range seismic signal recording;
- Configurable sample rate and preamplifier gain;
- High sensitive GNSS module equipped with active patch antenna supports GPS, Beidou, and GLONASS;
- Intermittent GNSS clock synchronization strategy, with timing accuracy less than $\pm 10\mu s$;
- Internal lithium-ion battery pack supports seismic recording for up to 1200 hours;
- Built-in test signal generator supports on-site full signal chain function and performance self-inspection and diagnosis, including system noise, real-time dynamic range, geophone impedance, natural frequency, sensitivity, damping, distortion, etc.;
- BLE module together with visual LEDs for onsite configuration and proximity. Ground communication distance around 20 meters, air communication distance over 100 meters, supports vehicle or UAV patrol;
- P&R (Place & Record) mode for rapid and convenient infield deployment, supporting automatic GNSS position matching with line and point number;
- Industrial Android tablet supports working status verification after deployment, including battery autonomy, environmental temperature, tilting, ambient noise, real-time seismic waveform, etc.;
- High strength engineering plastic enclosure for enhanced anti-dropping capability, waterproof (IP68), flame retardant (5VA), ultraviolet aging (F1);
- Compact volume (9.8L x 10.8W x 11H cm) and light weight (750g) for convenient transportation and deployment.



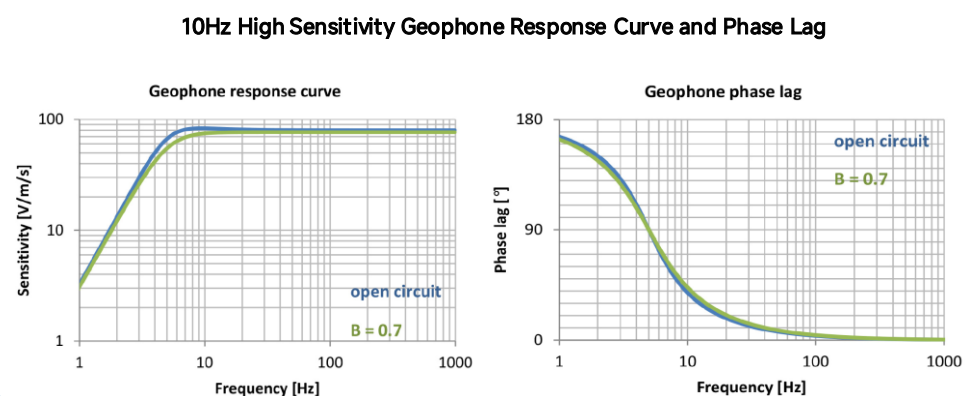
10Hz High Sensitivity Geophone

Natural Frequency	10Hz
Sensitivity	85.8V/m/s
Spurious	>240Hz
Distortion	≤0.1%
Coil Resistance	1800Ω
Tilt	±20°



5Hz High Sensitivity Geophone

Natural Frequency	5Hz
Sensitivity	80V/m/s
Spurious	>150Hz
Distortion	≤0.1%
Coil Resistance	1850Ω
Tilt	±10°



Standard Download/Charging Rack

Slots	32
Charging Time	3.5 hours
Charging Power	700W
Unit Charging Power	18W
Full Download Rate	18GB/min.
Unit Download Rate	0.6GB/min.
Interface	10G SFP+
Operating Temperature	0°C ~ +40°C
Size	194x64x38cm

10Hz High Sensitivity Geophone Response Curve and Phase Lag



Portable Download Unit



Portable Download Rack



Portable Charging Rack



Desktop Download/Charging Rack

Slots	14
Charging Time	3.5 hours
Charging Power	3500W
Unit Charging Power	18W
Full Download Rate	8GB/min.
Unit Download Rate	0.6GB/min.
Interface	2.5G Ethernet
Operating Temperature	0°C ~ +40°C
Size	94x64x38cm



SDM Server



Integrated Download/Charging Rack